VALKYRIES, BALLHALLA

The Valkyrie Fast Break

A basketball-themed digestive-system marble coaster.

STUDENT NAME

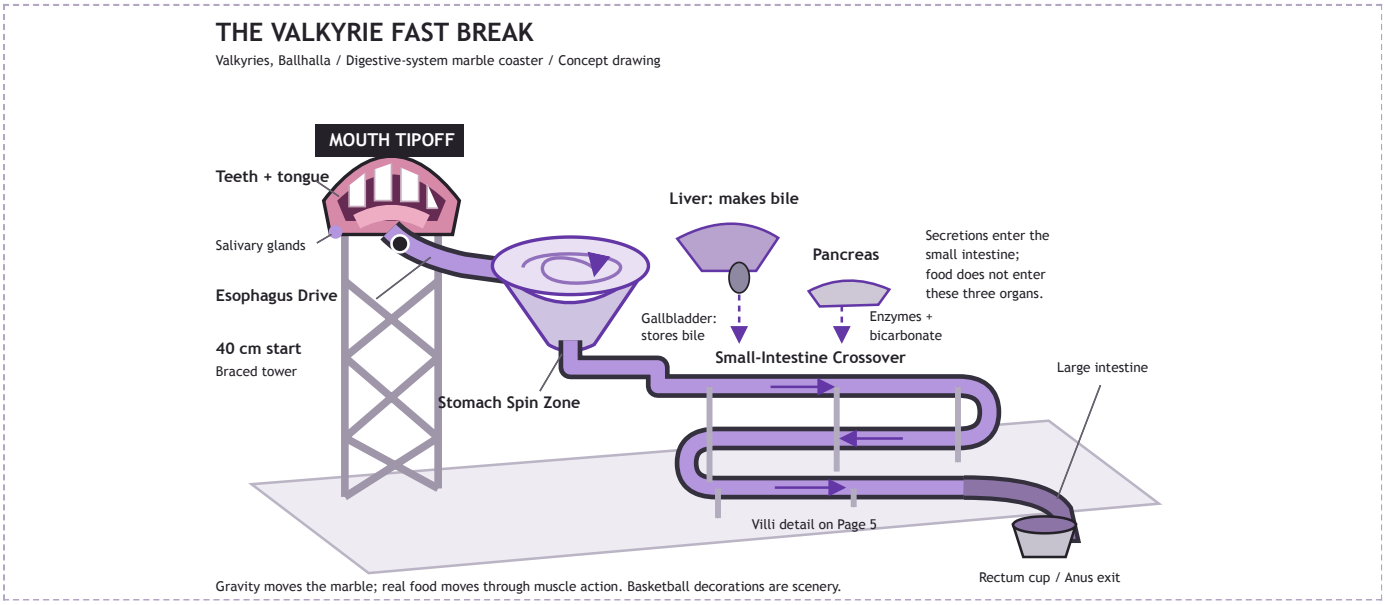
Piya Patel

TEACHER NAME

Miss.Narveson

Valkyries, Ballhalla	25 acres	Ages 8-14
Fictional park setting	Imagined park size	Fans, families & school groups

Valkyries Adventure Park is an imaginary 25-acre science theme park in the setting of Valkyries, Ballhalla, inspired by the Golden State Valkyries WNBA basketball team. Violet, black, white, and silver connect its basketball games, body-system rides, and arena-style signs. Designed for ages 8-14, families, and school groups, the park includes Fuel the Fast Break Cafe, water-refill stations, accessible paths, and courtside rest areas. Its featured ride, The Valkyrie Fast Break, follows a pretend bite of food through a mouth-tipoff tower, an esophagus descent, a stomach swirl, and intestine turns. A rolling marble demonstrates the route, while typed organ labels explain how digestion supplies nutrients the body can use.



From Tipoff To The Final Buzzer

The Valkyrie Fast Break is an imaginary basketball-themed roller coaster through the digestive system. Riders become a pretend bite of food, not a basketball that the body actually swallows. At Mouth Tipoff, white teeth and a pink tongue introduce mechanical digestion, while signs explain that salivary amylase begins breaking down starch. Imaginary passengers board a Courtside Bolus Buggy and descend the Esophagus Drive, where narration describes peristalsis, the muscular waves that push food forward. Arena-style cheers give way to low rumbling inside the Stomach Spin Zone. Here, churning, stomach acid, and enzymes help turn food into chyme. The Small-Intestine Crossover winds past giant villi, which increase the surface area for nutrient absorption. Displays explain that the liver makes bile, the gallbladder stores it, and the pancreas supplies enzymes and bicarbonate that helps neutralize acidic chyme. Food does not travel through these accessory organs. Most absorbed sugars and amino acids enter blood capillaries; many absorbed fats first enter lacteals in the lymphatic system. The Large-Intestine Home Stretch explains absorption of remaining water before the route reaches a rectum station and anus-themed exit. In the classroom model, a smooth marble follows this route under gravity and stops in a catch cup. Basketball scenery celebrates the team theme, but real digestion uses muscles and chemical processes, and food does not stay whole like a marble.

Science Along The Track

Mechanical digestion
Physical breakdown and mixing of food.

Chemical digestion
Enzymes break large food molecules into smaller ones.

Peristalsis
Muscular waves move contents through the digestive tract.

Chyme
The mixture produced as the stomach mixes food and digestive juices.

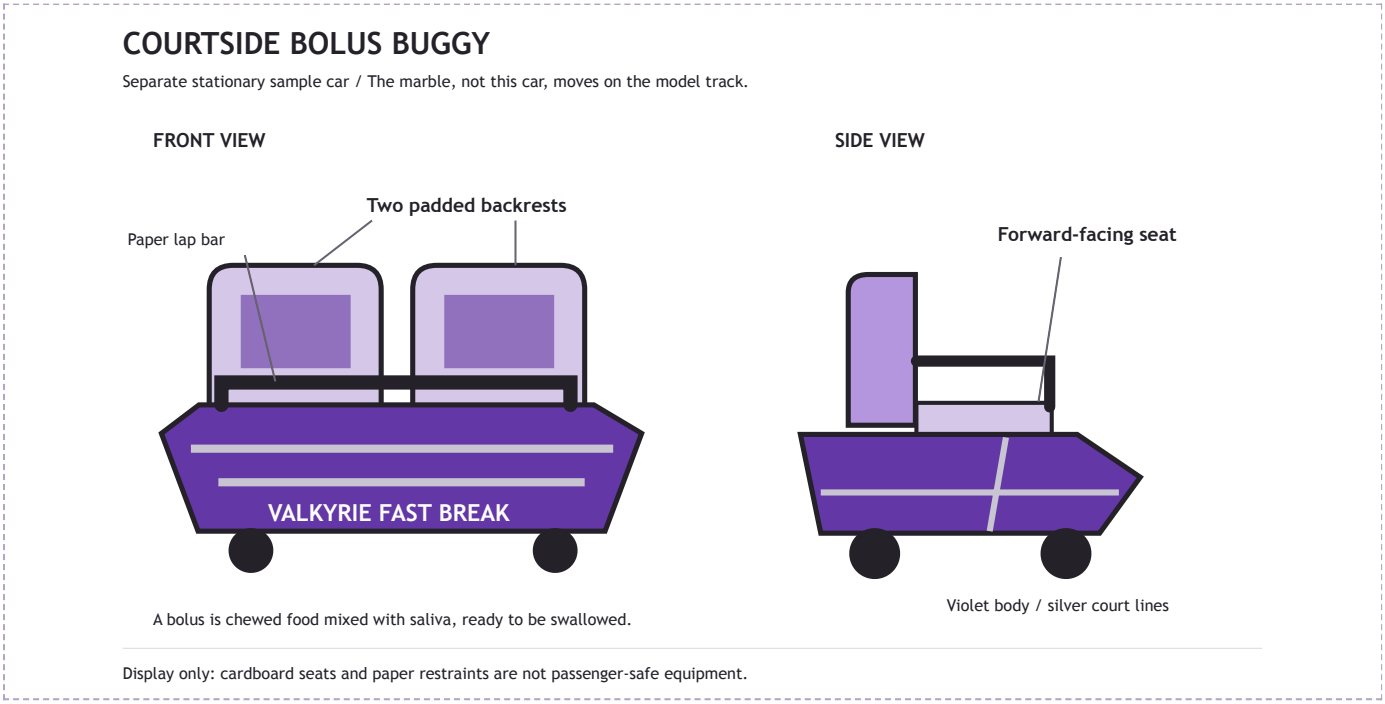
Villi
Fingerlike structures that increase absorption surface area.

Absorption
Nutrients cross the intestinal lining into blood or lymph.

> The sound effects, passenger cars, restraints, and braking belong to the imaginary full-size ride. The handmade model demonstrates the route with a marble; it is not a passenger-safe ride design.

The Courtside Bolus Buggy

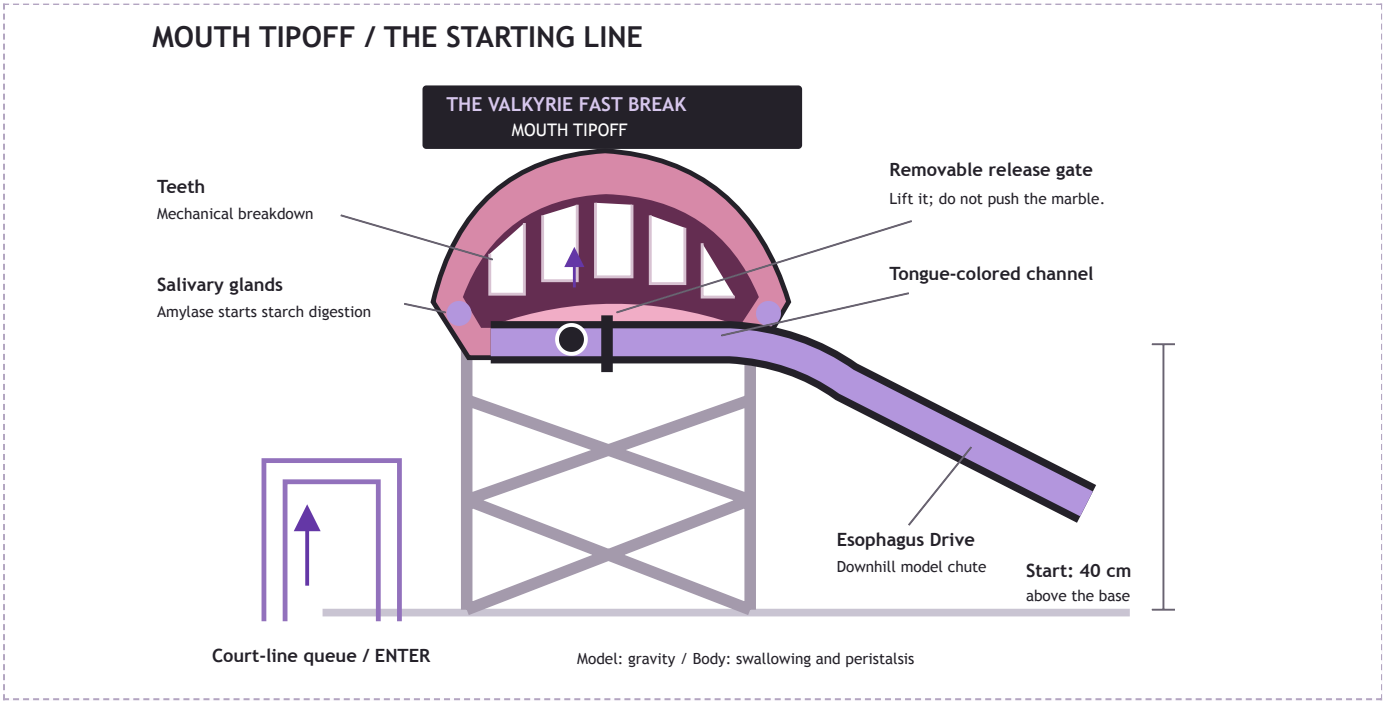
The imaginary Courtside Bolus Buggy has two padded seats side by side, with backrests and forward-facing views. Its violet body, black court-line patterns, and silver accents match the Valkyries-inspired park. A bolus is chewed food mixed with saliva ready to be swallowed, so the name connects the ride to digestion. Riders imagine settling into their seats and hearing the lap restraints click before tipoff. The full-size story includes engineered restraints and controlled braking; these are not working passenger equipment in the model. A separate stationary cardboard car displays two seats and a paper lap bar. The smooth marble, not this sample car, rolls along the track.



> **Seat requirement:** Confirm with the teacher that a separate sample car and seat illustration satisfy this requirement for the marble-run model.

Mouth Tipoff

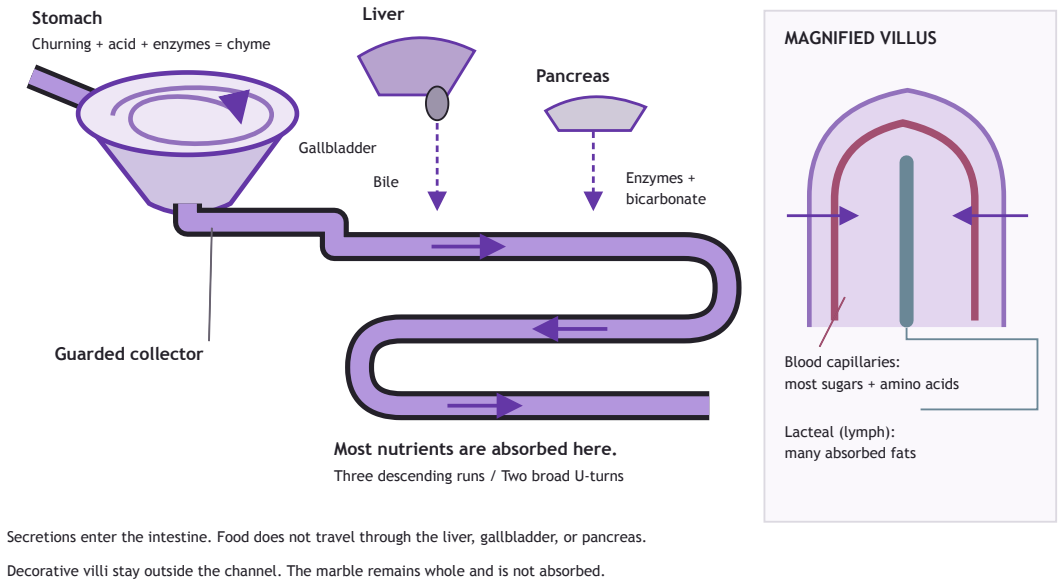
Mouth Tipoff is a giant pink mouth arch framed by violet-and-silver arena-style signs. White teeth rise above a tongue-colored loading channel on a braced tower, and a scoreboard-shaped sign reads The Valkyrie Fast Break. Queue rails resemble the lines around a basketball court. In the imaginary park, a whistle and cheering crowd announce tipoff as riders feel anticipation before the descent. A sign explains that teeth begin mechanical digestion and salivary amylase starts starch digestion. On the model, I place one marble behind a removable cardboard gate about 40 cm above the base, then lift the gate without pushing. Gravity supplies the model's motion; the mouth scenery does not chew or propel the marble.



> Height is a model-design choice, not an anatomical measurement. The teeth and tongue are scenery and do not chew or push the marble.

Spin Zone & Crossover Turns

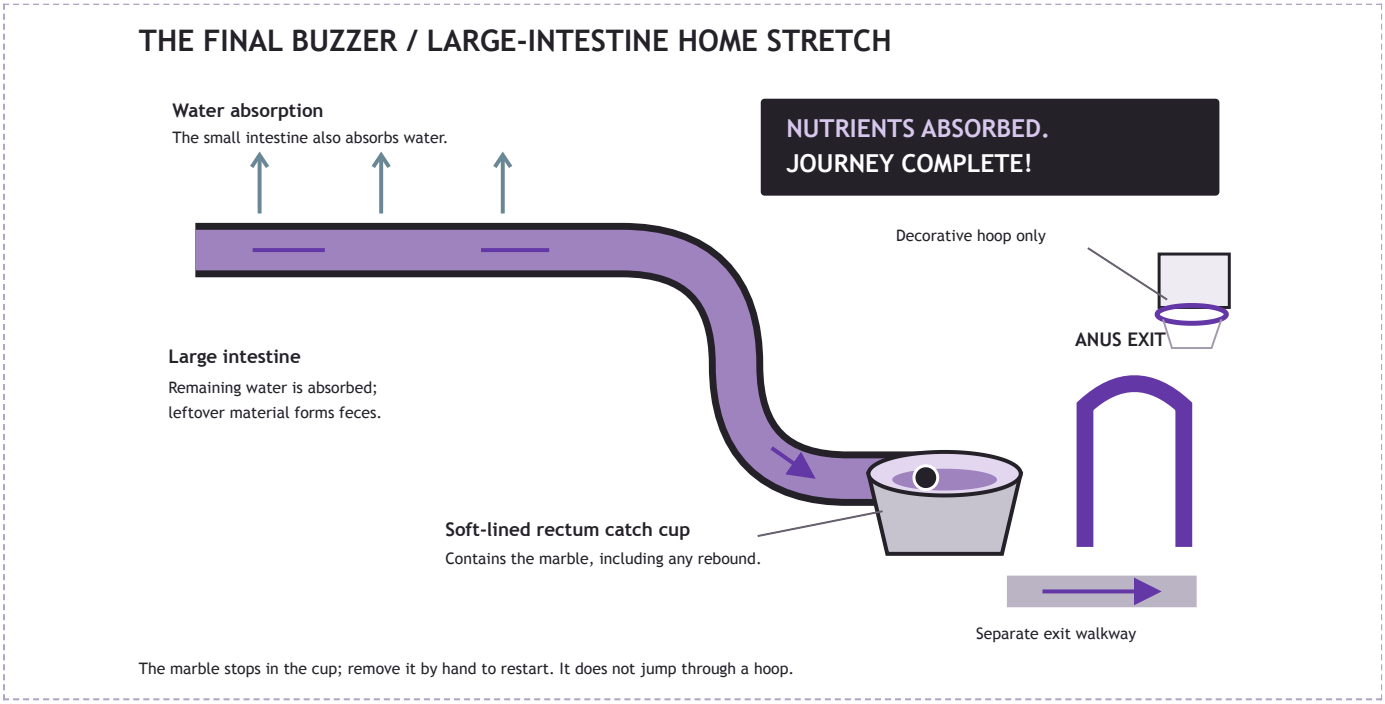
Halfway through the adventure, riders sweep into the Stomach Spin Zone, where low rumbling replaces the crowd noise. Swirling patterns suggest churning before the car enters the Small-Intestine Crossover, a series of winding turns named after a basketball change of direction. Imaginary passengers feel the car curve from side to side and see giant villi beside the route. Narration explains that these fingerlike structures increase the surface area for nutrient absorption, which happens mostly in the small intestine. Liver, gallbladder, and pancreas displays connect to the intestine with secretion arrows, not passenger tracks. On the model, the marble may circle inside an open funnel, pass through a guarded collector, and follow three descending runs joined by two broad turns. Its speed depends on slope, friction, and release; an apparent pause or fast turn is not a measurement of real digestion. Paper villi remain outside the rolling channel.



> Nutrient arrows point out of the small intestine toward blood or lymph. The marble remains on the ride because it is a simplified route marker.

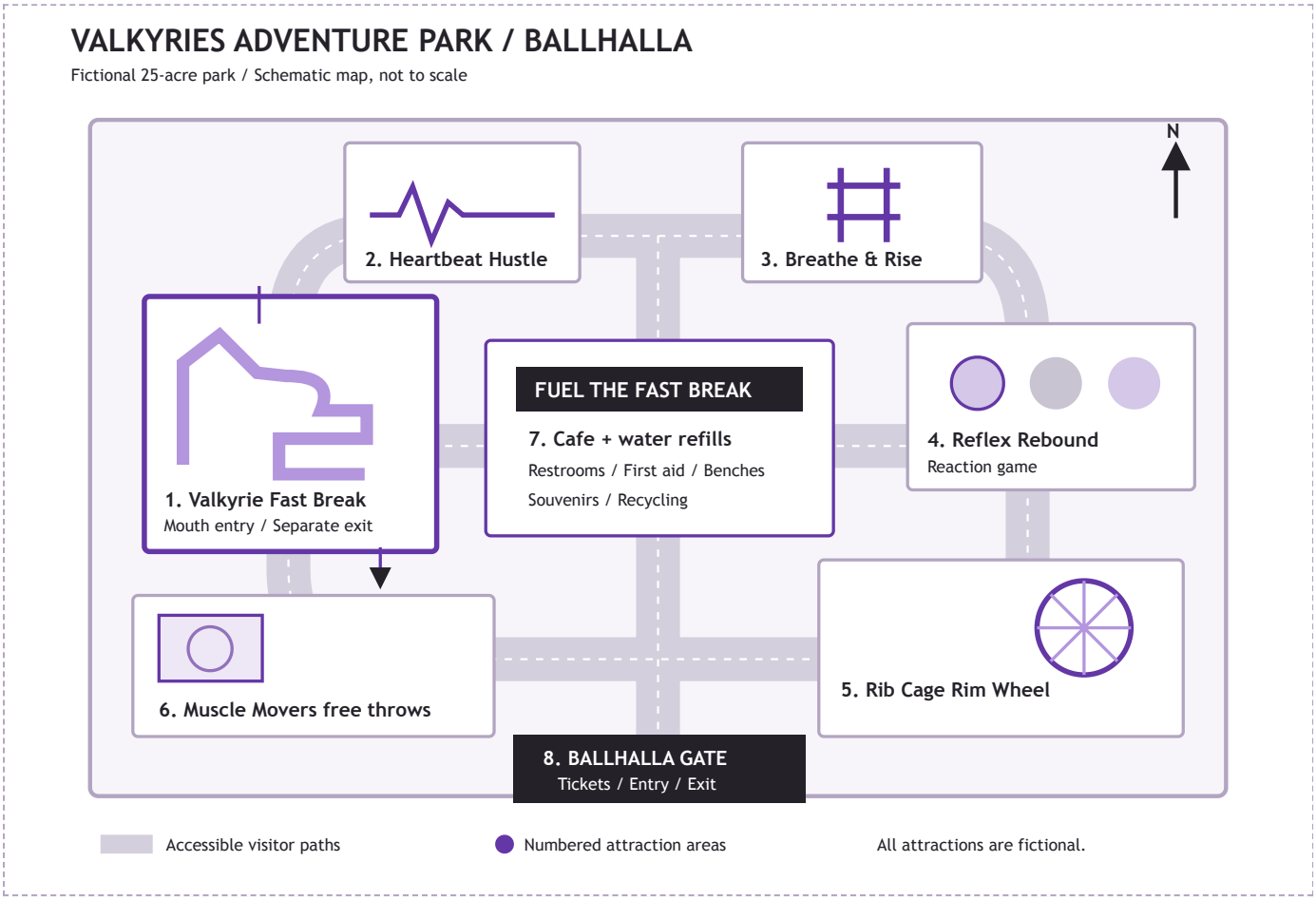
The Final Buzzer

The Large-Intestine Home Stretch curves past blue arrows showing absorption of remaining water; a sign notes that the small intestine also absorbs water. In the imaginary attraction, the rumbling fades and a final-buzzer sound announces the approach to the unloading station. A narrator explains that leftover material forms feces, the rectum stores it, and the anus is the final opening. Riders feel a gentle controlled stop and see a scoreboard reading, "Nutrients absorbed. Journey complete!" On the model, the marble rolls into a soft-lined catch cup representing the rectum station, beside an anus-themed exit arch and a decorative basketball hoop. The hoop is scenery: the marble does not jump through it or leave the guarded track. I remove the marble by hand to restart. Real food is broken down and partly absorbed, unlike the intact marble. A separate exit walkway returns visitors to Ballhalla.



> The catch cup must contain the marble, including any rebound. Real test results must be recorded during construction; this portfolio does not claim that the proposed model has already been tested.

Around Ballhalla



West: The Valkyrie Fast Break

Northwest: Heartbeat Hustle coaster

Northeast: Breathe & Rise drop tower

East: Reflex Rebound reaction game

Southeast: Rib Cage Rim Wheel

Southwest: Muscle Movers free-throw game

Center: Fuel the Fast Break Cafe & hydration

South: Ballhalla gate & ticket booths

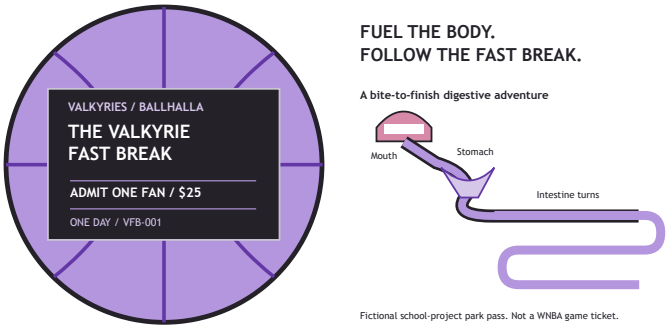
Also include first aid, restrooms, courtside benches, recycling, a souvenir stand, and accessible paths. Keep The Valkyrie Fast Break's entrance, exit, and route consistent with the physical model. All listed attractions are fictional school-project designs.

Your Ballhalla Game-Day Pass

VALKYRIES / BALLHALLA **VFB-001**

The Valkyrie Fast Break

Fuel the body. Follow the fast break.



Computer-generated illustration. A basketball-shaped pass in violet, black, white, and silver, with the ride name, admission price, and miniature digestive route.

ADMIT ONE FAN **ONE-DAY PARK PASS** **Fictional price: $25**

My basketball-shaped ticket celebrates the Golden State Valkyries-inspired theme while showing the mouth entrance and winding digestive route. Violet, black, white, and silver connect it to the rest of the park. This is a fictional school-project park pass, not an official WNBA or Golden State Valkyries game ticket. Its price and number are invented.

Peristalsis / Chyme / Villi / Absorption

Ticket-back challenge: Where are most nutrients absorbed?
Answer: The small intestine.

Science References

NIDDK: Your Digestive System & How It Works
KidsHealth: Your Digestive System